
Partial Surface Point Cloud
$P_H = (p_1, \ldots, p_N)$
Previous camera poses
$H = \{c_j, j \leq t\}$
First Neural Module
Input:
-Partial Surface PC: P_H
-Proxy Points: x_i
Output:
-Occupancy Probabilities: σ_i
Proxy Points and Occupancy Probabilities
$(x_1, \sigma_1), \ldots, (x_M, \sigma_M)$
Occupancy Probability
0.5
1
Second Neural Module
Input:
-Proxy Points & Occ.: (x_i, σ_i)
-Camera History Features: $h_H(x_i)$
Output:
-Spherical Mappings of Visibility Gains: ϕ_i
Proxy Points and Visibility Gains
$(x_1, \phi_1), \ldots, (x_M, \phi_M)$
Visibility Gain in direction $c_{t+1} - x$
0
1
New camera pose
c_{t+1}
c_t
True Surface
Proxy points x_i
c_t
True Surface
c_{t+1}
$c_{t+1} - x_i$
$c_{t+1} - x_j$
Proxy points x_i